Supplementary Material for Residual Corrected Flow for Controllable Human-Centric Editing Across Video Shots and 3D Assets

Wenshuo Gao, Jiangyue Zeng, and Shuai Yang, *Member, IEEE,*

This supplementary material provides the full algorithms and additional visual results. Appendix A describes condition preparation and inference for video and native 3D editing. Appendix B examines backbone generalization and residual reusing. Appendices C and D present more video and native 3D editing results, respectively. Appendix E provides additional qualitative comparisons.

APPENDIX A
FULL ALGORITHM

This section describes condition preparation and inference for both branches. We write the updates in latent space and omit the internal operations of the pretrained encoders and decoders. Masks are resized or mapped to the corresponding latent coordinates before they are used.

A. Video Conditions

Given a source video X, source and target descriptions $q_{\mathrm{src}}, q_{\mathrm{tar}}$, and an edit instruction p, we prepare appearance and structural conditions separately. FASHN Human Parser initializes person and face/hair masks from the first frame, and MatAnyone propagates the two masks independently, giving the person mask P and head mask H throughout the video.

a) Source and target appearance: The source reference r_{src} is the first video frame. The image editor produces $r_{\mathrm{tar}} = G(r_{\mathrm{src}}, p, \mathcal{A})$, where $\mathcal{A}$ contains any enabled background, clothing, or object references and may be empty. The instruction identifies the role of each supplied image.

b) Shared structural control: Source depth D, Canny edges C, and HED boundaries E form the regional control

$$S = H \odot (w_D D + w_C C + w_E E) + (P - H) \odot D.$$

We use $w_D = 0.8$ and $w_C = w_E = 0.1$. The head therefore receives depth and edge information, the remaining person receives depth, and the background receives no structural constraint. Both branches use the same S, with conditions $c_{\mathrm{src}} = (q_{\mathrm{src}}, r_{\mathrm{src}}, S)$ and $c_{\mathrm{tar}} = (q_{\mathrm{tar}}, r_{\mathrm{tar}}, S)$.

B. Video Inference

We encode X into z_0 and sample one Gaussian noise ϵ for the entire trajectory. The known source path is $z_t = (1-t)z_0 + t\epsilon$, with denoising velocity $V_0 = z_0 - \epsilon$. At each refresh step t_i, the source prediction gives $V^{\mathrm{res}}_{t_i} = V_0 - V^{\mathrm{src}}_{t_i}(z_{t_i})$. Between refreshes, $V^{\mathrm{res}}_{t_i}$ stores the last computed residual, while the target velocity is evaluated at every step. We use a reuse interval of $r = 10$. With one target prediction per step and one source prediction every r steps, this requires only $1 + 1/r = 1.1$ times the flow evaluations of pure generation under the same guidance settings. Applying the residual within H preserves identity-bearing content without imposing the same reconstruction constraint on clothing or background:

$$V^{\mathrm{edit}}_{t_i}(z^{\mathrm{edit}}_{t_i}) = V^{\mathrm{tar}}_{t_i}(z^{\mathrm{edit}}_{t_i}) + H \odot V^{\mathrm{res}}_{t_i}.$$

After each update, source high frequencies are transferred within the head region. Let LF and HF denote the low- and high-frequency components. The transfer operator is

$$\mathcal{T}_H(u, z) = u + \lambda H \odot [\mathrm{HF}(z) - \mathrm{HF}(u)].$$

We use a frequency threshold of 0.8 for the low/high-frequency split and set the source high-frequency weight to $\lambda = 0.5$. Outside H, the operator leaves the edited latent unchanged. The source latent is taken at the same noise level as the updated target. Algorithm S1 gives the complete inference procedure.

C. Native 3D Conditions

The supplied mesh M is encoded into shape features z_0^{sh} and appearance features z_0^{tex} on a common sparse coordinate set $\mathcal{C}$. Source views supply the appearance condition, and the image editor constructs a target view from the source frontal view, edit instruction, and optional clothing reference. The source and target images use a shared source-derived crop and are separately masked to isolate the person.

a) Spatial head mask: Face and hair labels from the source views are projected onto the observed surface. These labels are associated with the sparse latent coordinates to obtain $H_{\mathcal{C}}$. Both editing stages use this source-derived spatial mask, so preservation is defined on the asset rather than separately in each rendered view.

b) Shape and appearance conditions: The shape stage uses the source and target image conditions to edit geometry. The appearance stage additionally receives shape features: source shape conditions the source branch, while the edited

Algorithm S1 Residual-Corrected Flow for Video Editing

Require: Initial noise ϵ, source latent z_0, source and target conditions $c_{\mathrm{src}}, c_{\mathrm{tar}}$, schedule $0 = t_0 < \cdots < t_N = 1$, head mask H, reuse interval r, high-frequency weight λ
Ensure: Edited latent z_0^{edit}

$V_0 \leftarrow z_0 - \epsilon$; $z_{t_N}^{\mathrm{edit}} \leftarrow \epsilon$
for $i = N, N-1, \ldots, 1$ **do**
  $z_{t_i} \leftarrow (1 - t_i) z_0 + t_i \epsilon$
  $z_{t_{i-1}} \leftarrow (1 - t_{i-1}) z_0 + t_{i-1} \epsilon$
  if $(N - i) \bmod r = 0$ **then**
    $V_{t_i}^{\mathrm{src}}(z_{t_i}) \leftarrow F_\theta(z_{t_i}, t_i, c_{\mathrm{src}})$
    $V_{t_i}^{\mathrm{res}} \leftarrow V_0 - V_{t_i}^{\mathrm{src}}(z_{t_i})$
  else
    $V_{t_i}^{\mathrm{res}} \leftarrow V_{t_{i+1}}^{\mathrm{res}}$
  end if
  $V_{t_i}^{\mathrm{tar}}(z_{t_i}^{\mathrm{edit}}) \leftarrow F_\theta(z_{t_i}^{\mathrm{edit}}, t_i, c_{\mathrm{tar}})$
  $V_{t_i}^{\mathrm{edit}} \leftarrow V_{t_i}^{\mathrm{tar}}(z_{t_i}^{\mathrm{edit}}) + H \odot V_{t_i}^{\mathrm{res}}$
  $z_{t_{i-1}}^{\mathrm{edit}} \leftarrow z_{t_i}^{\mathrm{edit}} + (t_i - t_{i-1}) V_{t_i}^{\mathrm{edit}}$
  $z_{t_{i-1}}^{\mathrm{edit}} \leftarrow \mathrm{LF}(z_{t_{i-1}}^{\mathrm{edit}}) + \lambda H \odot \mathrm{HF}(z_{t_{i-1}}) + (1 - \lambda H) \odot \mathrm{HF}(z_{t_{i-1}}^{\mathrm{edit}})$
end for
return z_0^{edit}

shape conditions the target branch. Thus, material editing follows the edited geometry while source reconstruction remains tied to the supplied asset.

D. Native 3D Inference

Shape and appearance are edited sequentially. Within each stage, z_0 denotes the source representation, and $V_{t_i}^{\mathrm{src}}$ and $V_{t_i}^{\mathrm{tar}}$ denote the stage-specific fields under its source and target conditions. We use the same denoising convention as the main paper: $V_0 = z_0 - \epsilon$, and the update from t_i to t_{i-1} adds $(t_i - t_{i-1}) V_{t_i}$.

The residual $V_{t_i}^{\mathrm{res}} = V_0 - V_{t_i}^{\mathrm{src}}(z_{t_i})$ corrects the source reconstruction error. To apply frequency transfer, we first form the unmasked residual-corrected proposal $\widetilde{z}_{t_{i-1}}^{\mathrm{edit}}$ and its difference Δ_i from the known source latent $z_{t_{i-1}}$. The spatial head mask then blends the filtered correction with the ordinary target step. This reduces to the masked velocity update:

$$V_{t_i}^{\mathrm{edit}}(z_{t_i}^{\mathrm{edit}}) = V_{t_i}^{\mathrm{tar}}(z_{t_i}^{\mathrm{edit}}) + H_{\mathcal{C}} \odot V_{t_i}^{\mathrm{res}}.$$

Frequency transfer uses the actual coordinates $\mathcal{C}$, with the occupancy-normalized operator $\mathrm{LP}_{b,\mathcal{C}}$ defined in Eq. (13). As in the main paper,

$$\mathcal{T}(\Delta_i) = \mathrm{LP}_{b,\mathcal{C}}(\Delta_i) + \eta[\Delta_i - \mathrm{LP}_{b,\mathcal{C}}(\Delta_i)].$$

We use filtering strength $b = 0.2$ and retained high-frequency fraction $\eta = 0.5$ in both stages. The source prediction is recomputed at each 3D step. Algorithm S2 gives the procedure; the stage variable s selects the shape or appearance model and its corresponding conditions.

Algorithm S2 Residual-Corrected Flow for Native 3D Editing

Require: Source latents $z_0^{\mathrm{sh}}, z_0^{\mathrm{tex}}$, source and target image conditions, schedule $0 = t_0 < \cdots < t_N = 1$, spatial head mask $H_{\mathcal{C}}$, frequency-transfer operator $\mathcal{T}$
Ensure: Edited latents $z_0^{\mathrm{sh,edit}}, z_0^{\mathrm{tex,edit}}$

for $s \in \{\mathrm{sh}, \mathrm{tex}\}$, in this order **do**
  $z_0 \leftarrow z_0^s$; form stage conditions $c_{\mathrm{src}}^s, c_{\mathrm{tar}}^s$
  if $s = \mathrm{tex}$ **then**
    Use z_0^{sh} in the source branch and $z_0^{\mathrm{sh,edit}}$ in the target branch
  end if
  Sample fixed Gaussian noise ϵ; $V_0 \leftarrow z_0 - \epsilon$; $z_{t_N}^{\mathrm{edit}} \leftarrow \epsilon$
  for $i = N, N-1, \ldots, 1$ **do**
    $z_{t_i} \leftarrow (1 - t_i) z_0 + t_i \epsilon$
    $z_{t_{i-1}} \leftarrow (1 - t_{i-1}) z_0 + t_{i-1} \epsilon$
    $V_{t_i}^{\mathrm{src}}(z_{t_i}) \leftarrow F_\theta^s(z_{t_i}, t_i, c_{\mathrm{src}}^s)$
    $V_{t_i}^{\mathrm{tar}}(z_{t_i}^{\mathrm{edit}}) \leftarrow F_\theta^s(z_{t_i}^{\mathrm{edit}}, t_i, c_{\mathrm{tar}}^s)$
    $V_{t_i}^{\mathrm{res}} \leftarrow V_0 - V_{t_i}^{\mathrm{src}}(z_{t_i})$
    $\bar{z}_{t_{i-1}}^{\mathrm{tar}} \leftarrow z_{t_i}^{\mathrm{edit}} + (t_i - t_{i-1}) V_{t_i}^{\mathrm{tar}}(z_{t_i}^{\mathrm{edit}})$
    $\tilde{z}_{t_{i-1}}^{\mathrm{edit}} \leftarrow \bar{z}_{t_{i-1}}^{\mathrm{tar}} + (t_i - t_{i-1}) V_{t_i}^{\mathrm{res}}$
    $\Delta_i \leftarrow \tilde{z}_{t_{i-1}}^{\mathrm{edit}} - z_{t_{i-1}}$
    $z_{t_{i-1}}^{\mathrm{edit}} \leftarrow (1 - H_{\mathcal{C}}) \odot \bar{z}_{t_{i-1}}^{\mathrm{tar}} + H_{\mathcal{C}} \odot [z_{t_{i-1}} + \mathcal{T}(\Delta_i)]$
  end for
  $z_0^{s,\mathrm{edit}} \leftarrow z_{t_0}^{\mathrm{edit}}$
end for
return $z_0^{\mathrm{sh,edit}}, z_0^{\mathrm{tex,edit}}$

APPENDIX B
ADDITIONAL ANALYSIS

A. Generalization across Backbones

We examine backbone generalization for video and native 3D editing in Tables S1 and S2. The video comparison considers Wan2.1, Wan2.2, and HunyuanVideo, while the native 3D comparison considers TRELLIS.2 and Hunyuan3D 2.1. As shown, our method maintains robust performance across different architectures.

TABLE S1
VIDEO EDITING ACROSS BACKBONES.

Backbone	SC ↑	PA ↑
Wan2.1	<u>0.8310</u>	<u>0.305</u>
Wan2.2	**0.8413**	**0.334**
HunyuanVideo	0.7287	0.268

TABLE S2
NATIVE 3D EDITING ACROSS BACKBONES.

Backbone	SC ↑	PA ↑
TRELLIS.2	**0.889**	**0.315**
Hunyuan3D 2.1	<u>0.772</u>	<u>0.275</u>

B. Analysis of Residual Reusing

We examine the trade-off between editing quality and processing time by varying the residual reuse interval r. Table S3 summarizes Detail Consistency, Prompt Alignment, and processing time at different intervals. The results indicate that increasing the reuse step r significantly reduces inference time with minimal impact on quality.

TABLE S3
EFFECT OF RESIDUAL REUSING.

r	DC ↑	PA ↑	Time (s) ↓
1	**62.14**	**0.307**	314
5	<u>61.93</u>	0.290	195
10	61.89	<u>0.305</u>	<u>189</u>
20	61.10	0.294	**170**

APPENDIX C
MORE VIDEO EDITING RESULTS

We present more video editing results in Figs. S1–S4.

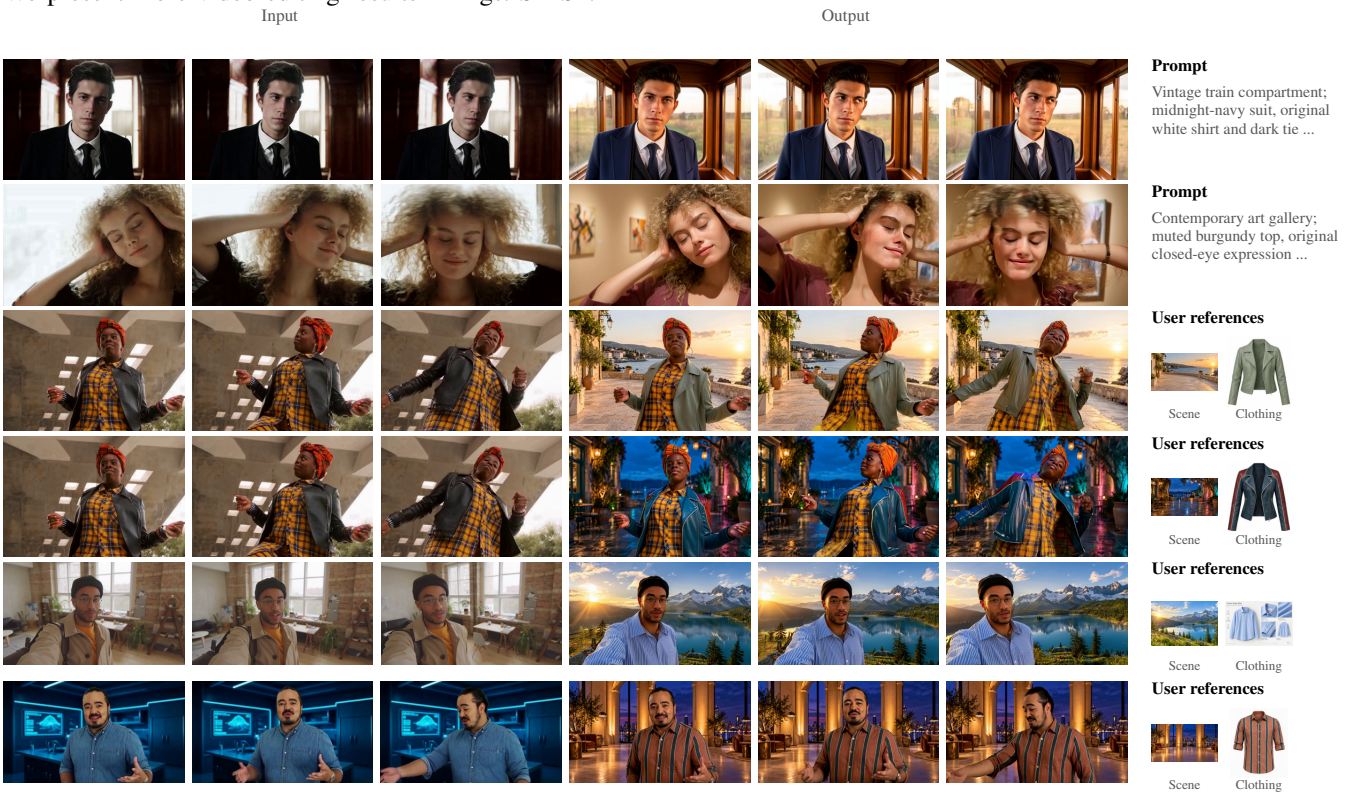


Fig. S1. More video editing results.

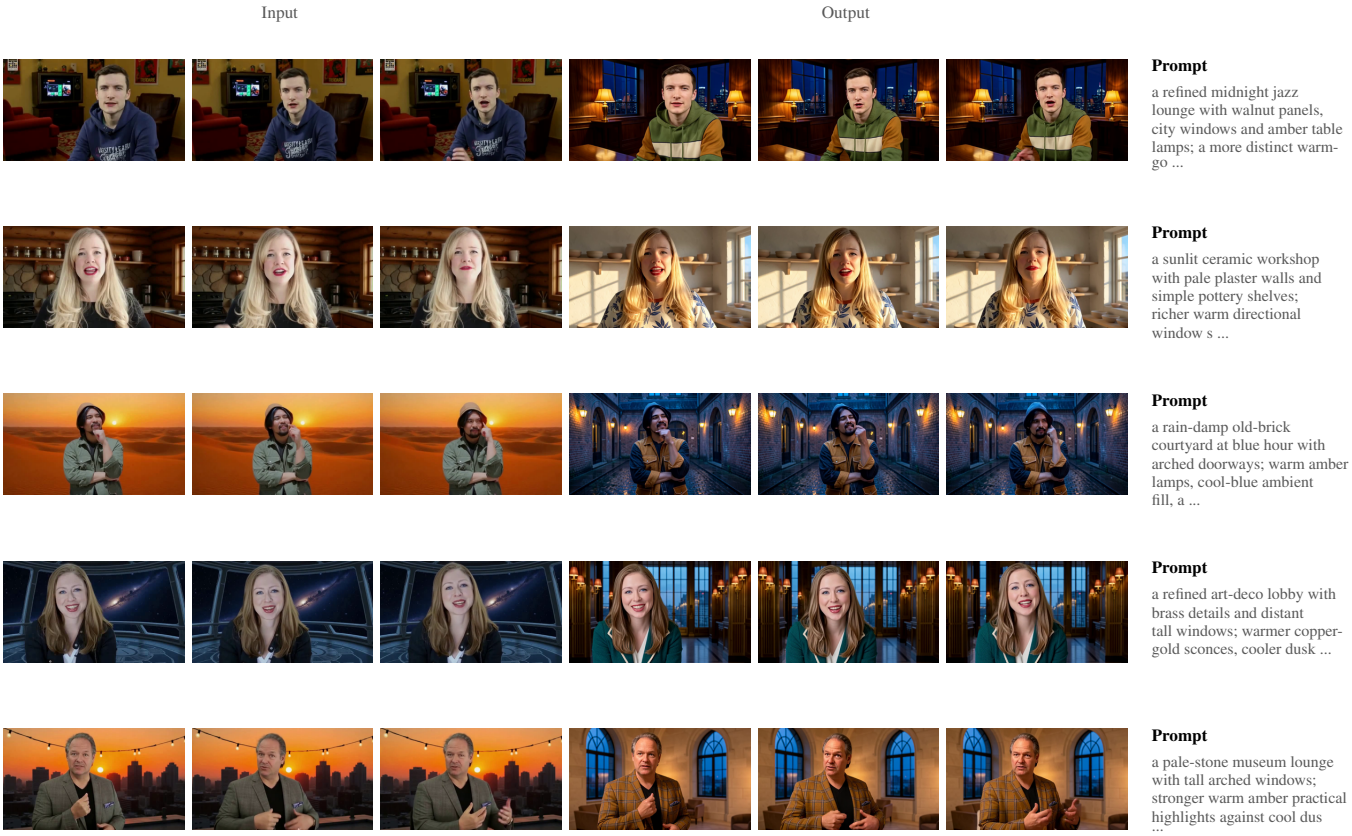


Fig. S2. More text-guided video editing results.

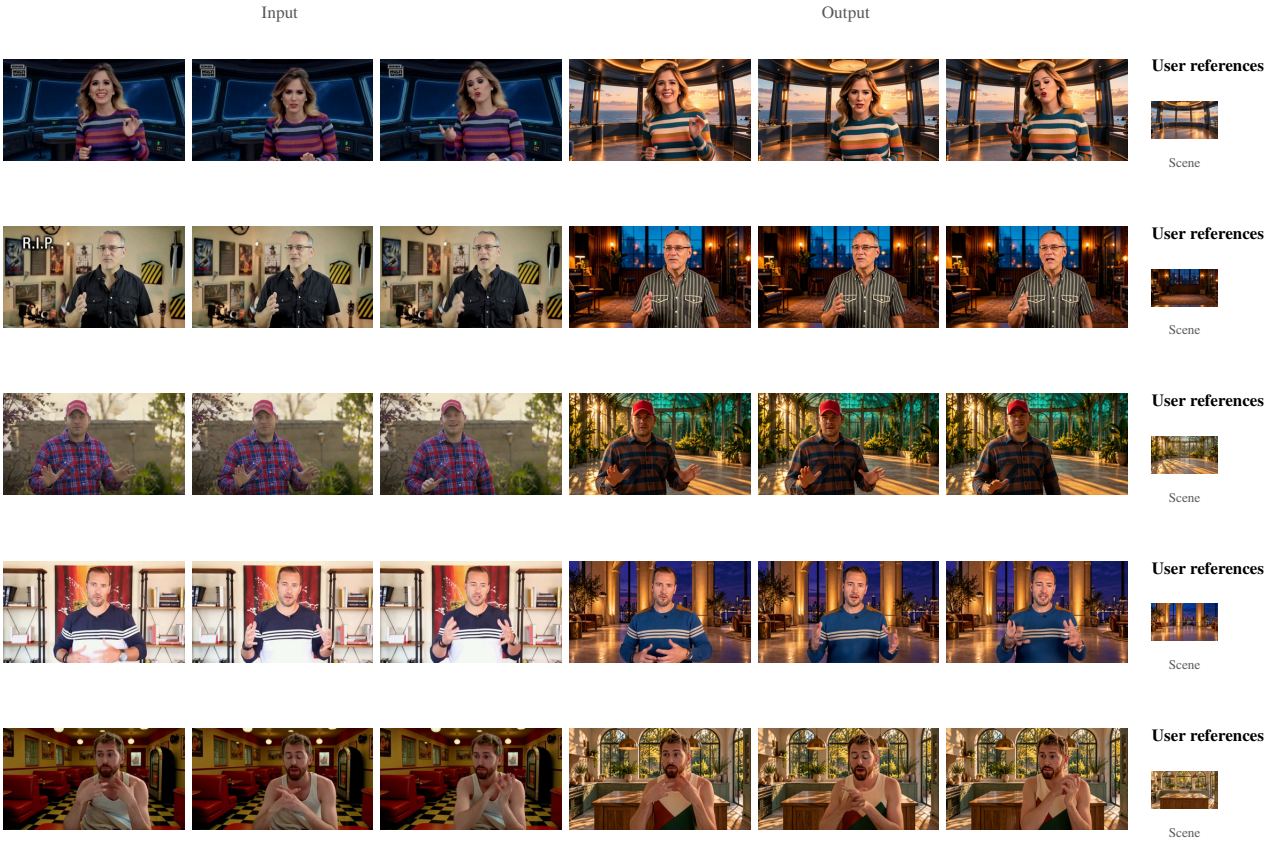


Fig. S3. More background-guided video editing results.

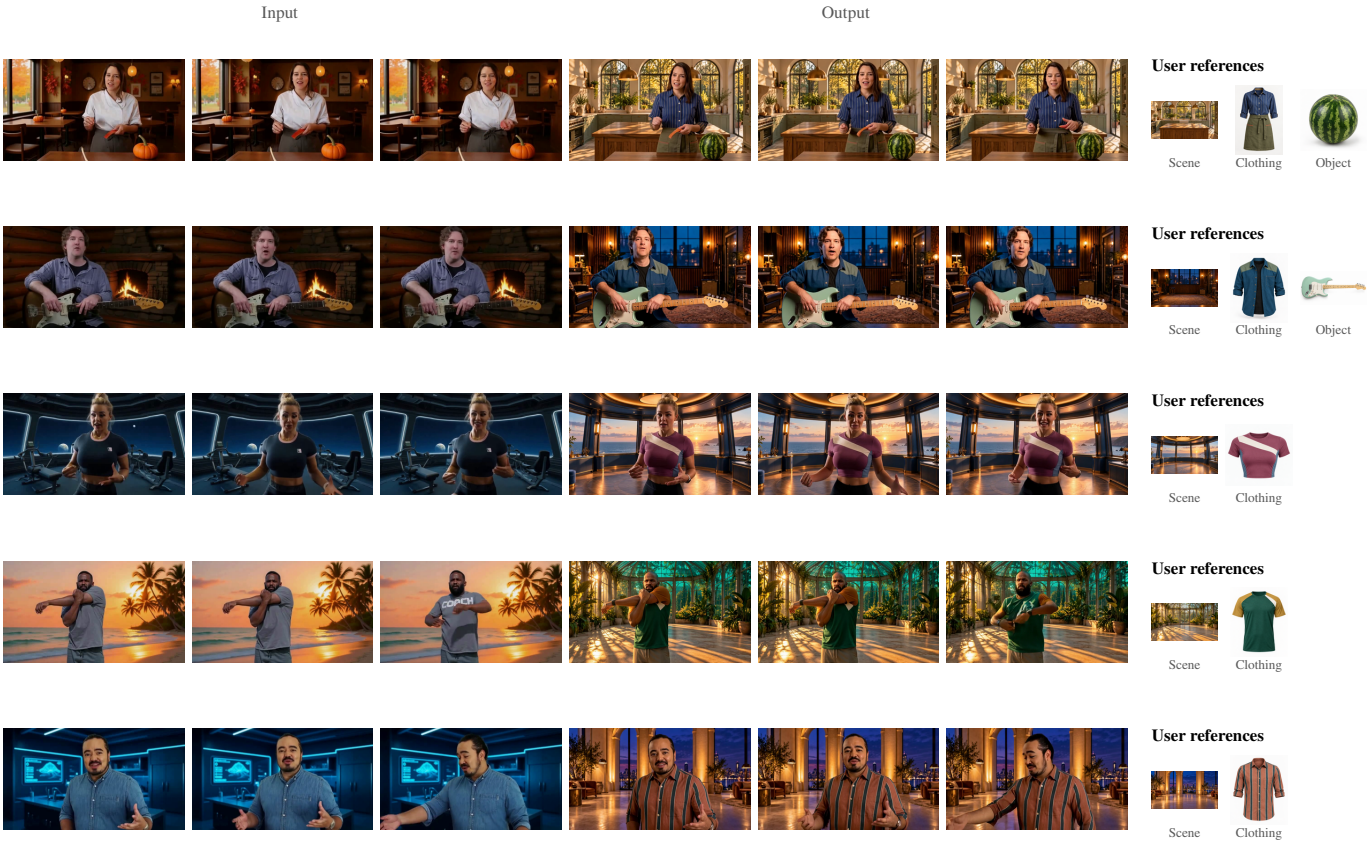


Fig. S4. More reference-guided video editing results.

APPENDIX D
MORE NATIVE 3D EDITING RESULTS

We present more native 3D editing results in Figs. S5 and S6.

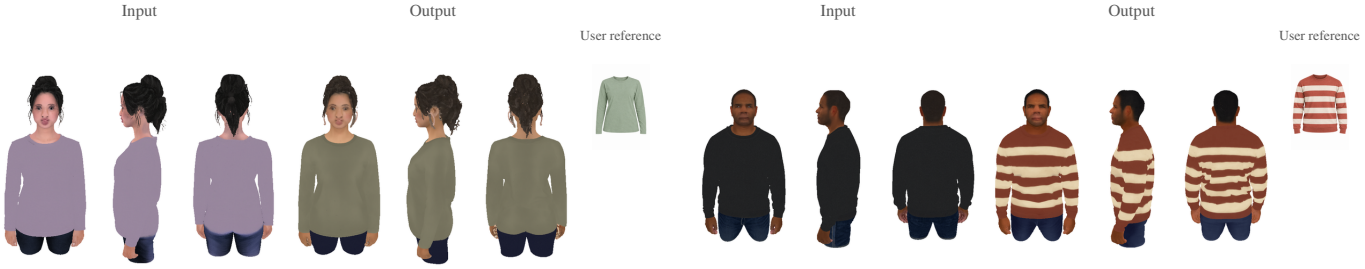


Fig. S5. More native 3D editing results.

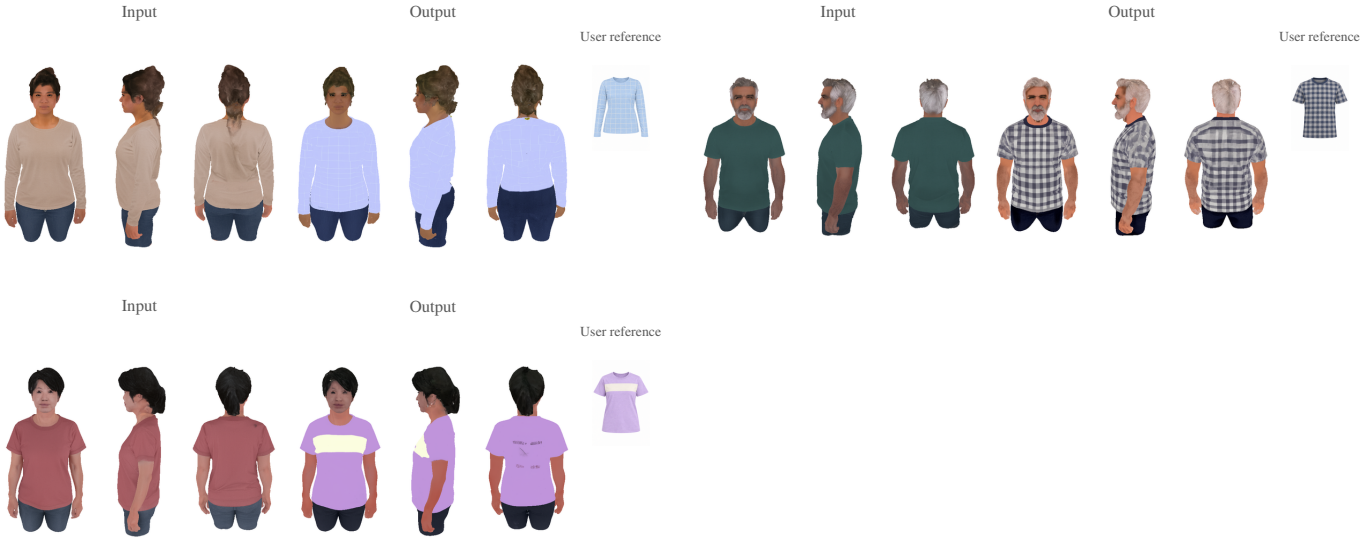


Fig. S6. Additional native 3D editing results.

APPENDIX E
MORE COMPARISONS

We present more qualitative video comparisons in Fig. S7.

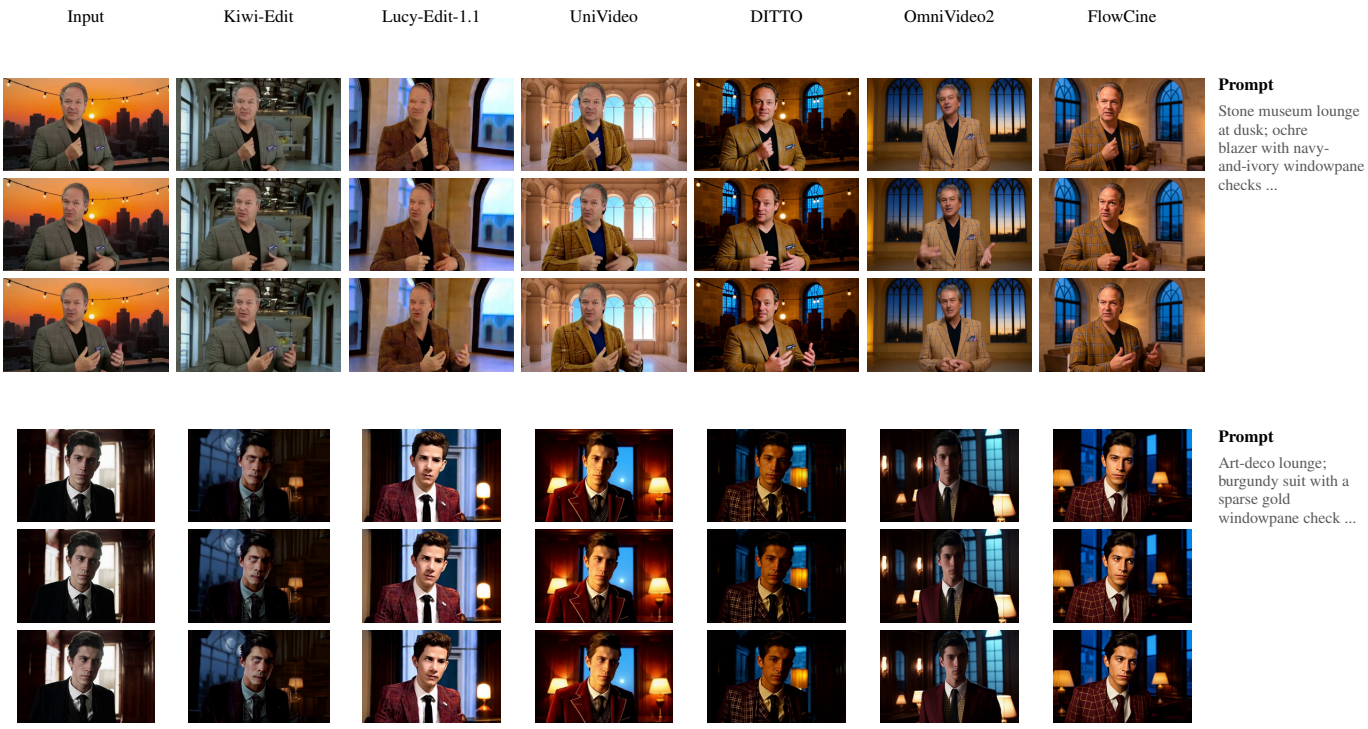


Fig. S7. More qualitative video comparisons.